

29. Which one of the following places has the lowest rainfall during the South-West Monsoon?

- (a) Kolkata (b) Mangalore
(c) Chennai (d) Delhi

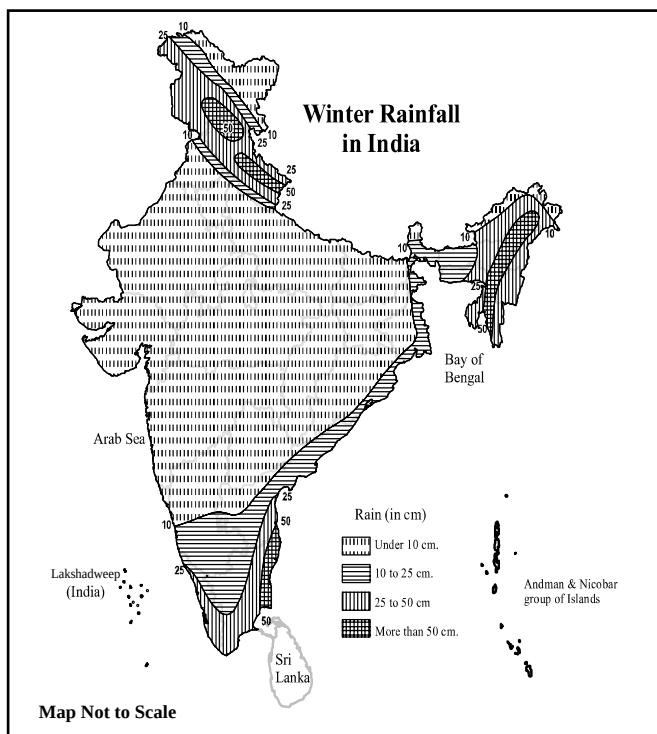
U.P.P.C.S. (Pre) 2016

Ans. (c)

The south-western monsoon is active from 1 June to 30 September. Maximum rainfall in India occurs due to this monsoon. Chennai receives the lowest rainfall from the south-western monsoon. Chennai receives maximum rainfall from the north-eastern monsoon.

iii. Winter Rainfall

*Winter monsoons do not cause rainfall as they move from land to the sea. Most parts of India **do not** receive rainfall in the winter season. Some weak temperate cyclones from the Mediterranean Sea cause rainfall in **Punjab, Haryana, Delhi** and Western **Uttar Pradesh**. Although the amount is meager, it is highly beneficial for **Rabi crops**. Central India, northern parts of the Southern Peninsula, Arunachal Pradesh and Assam also receive winter rainfall.



*During October and November, the **northeast monsoon** while crossing over the Bay of Bengal, picks up moisture

and causes torrential rainfall over the **Tamil Nadu coast, southern Andhra Pradesh, southeast Karnataka, and southeast Kerala.**

1. India's Northern plains get winter rainfall due to –

- (a) Western Disturbance
(b) Monsoon in Bay of Bengal
(c) Monsoon in Arabian sea.
(d) Retreating Monsoon.

U.P.P.C.S. (Pre) 1993

Ans. (a)

India's northern plains get winter rainfall due to a western disturbance. A western disturbance is an extratropical storm originating in the **Mediterranean region** that brings sudden winter rain to the northwestern part of the Indian subcontinent.

2. Which of the following regions of India receives winter rainfall?

- | | | |
|-----------------------|---|------------|
| (a) Odisha | – | Karnataka |
| (b) Punjab | – | Tamil Nadu |
| (c) Arunachal Pradesh | – | Bihar |
| (d) Tamil Nadu | – | Karnataka |

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

Coastal areas of Tamil Nadu receive winter rainfall of up to 65-75cm due to retreating North-East Monsoon. Additionally, Punjab also receives winter rainfall due to Western Disturbance.

3. Given below are two statements, one is labeled as Assertion (A) and other is Reason (R)-

Assertion (A) – The Tamil Nadu coast remains dry during South-West monsoon season.

Reason (R) – The Tamil Nadu coast is situated parallel to the Bay of Bengal branch of South-West monsoon and it lies in the rain shadow area of the Arabian Sea branch of the South-West monsoon during monsoon season.

Select the correct answer using the code given below :
Code :

- (a) (A) is false but (R) is true
(b) (A) is true but (R) is false
(c) Both (A) and (R) is true but (R) is not correct explanation of (A)
(d) Both (A) and (R) are true and (R) is correct explanation of (A)

U.P.P.C.S. (Pre) 2023

Ans. (d)

The Tamil Nadu coast does remain dry during the South-West monsoon season. This is because it is situated parallel to the Bay of Bengal branch of the South-West monsoon. The prevailing winds during the monsoon season blow from the South-West, bringing moisture-laden winds from the Bay of Bengal towards the Indian subcontinent. However, the Tamil Nadu coast lies in the rain shadow area of the Arabian Sea branch of the South-West monsoon. The Western Ghats mountain range acts as a barrier, blocking the moisture-laden winds from the Arabian Sea, resulting in less rainfall along the Tamil Nadu coast.

Therefore, both the assertion (A) and the reason (R) are true, and (R) provides the correct explanation for (A).

4. **Winter rainfall in Tamil Nadu is mostly due to –**

- (a) Western disturbances (b) South-West monsoon
(c) North-East monsoon (d) South-East monsoon

R.A.S./R.T.S. (Pre) 2010

U.P.P.C.S. (Pre) 2008

Ans. (c)

See the explanation of above question.

5. **Assertion (A) : Northern plains of India receive some rainfall in winter.**

Reason (R) : North East Monsoon is active in winter.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 1998

Ans. (b)

See the explanation of above question.

6. **Given below are two statements, in which one is labeled as Assertion (A) and the other as Reason (R) :**

Assertion (A) : During the winter season, most part of India is dry.

Reason (R) : Most parts of the country are under influence of the northeast trade winds during the winter season.

Select the correct answer using the code given below :

Code :

- (a) Both (A) and (R) are true and (R) is not the correct explanation of (A).
(b) (A) is false but (R) is true.

(c) (A) is true but (R) is false.

(d) Both (A) and (R) are true but (R) is the correct explanation of (A).

U.P. R.O./A.R.O. (Pre) 2023

Ans. (a)

During the winter season, northeast trade winds prevail over India. Winter Monsoons or the North-East Monsoons blow during the winter season. Notably these types of winds do not cause rainfall as they move from land to the sea. It is because, they have little humidity. Hence, most parts of India do not have rainfall in the winter season. It is important to mention here that, there is an exception, during October and November, northeast monsoon while crossing over the Bay of Bengal, picks up moisture and causes torrential rainfall over the southern state of Tamil Nadu.

7. **Of the following, the state which receives rainfall in winter season is**

- (a) Kerala (b) Tamil Nadu
(c) West Bengal (d) Orissa

44th B.P.S.C. (Pre) 2000

Ans. (b)

See the explanation of above question.

8. **Which of the following coastal region in India, receives rainfall from North Eastern monsoon ?**

- (a) Malabar Coast (b) Gujarat Coast
(c) Coromandel Coast (d) Konkan Coast

Jharkhand P.C.S. (Pre) 2021

Ans. (c)

In India, the Coromandel coastal region receives rainfall from the northeast monsoon. Apart from this, southeast Karnataka, and southeast Kerala also receive rainfall from the northeast monsoon.

9. **Which areas of India receive rainfall from North- East Monsoon?**

- A. Tamil Nadu Coast**
B. Gujarat Coast
C. Southern Andhra Pradesh
D. South-East Karnataka

Code :

- (a) A, B, C & D (b) A, C & D
(c) B, C & D (d) A, B & C
(e) Question not attempted

Raj. P.C.S. (Pre) 2023

Ans. (b)

See the explanation of the above question.

10. Assertion (A) : Anti-cyclonic conditions are formed in winter season when atmospheric pressure is high and air temperature is low.

Reason (R) : Winter rainfall in Northern India causes development of anticyclonic conditions with low temperature.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- (c) (A) is true, but (R) is false
- (d) (A) is false, but (R) is true

I.A.S. (Pre) 2001

Ans. (c)

Rainfall during winter in Northern India is due to the western disturbance. Anti-cyclonic conditions in India already exist, and they have nothing to do with winter rains. Anti-cyclonic conditions during winter are formed when the atmospheric pressure is high and the temperature is low. Therefore, Assertion (A) is correct and Reason (R) is wrong.

11. Which of the following areas of India receives winter rainfall ?

- (a) North-West
- (b) South-West
- (c) North-East
- (d) South-East

U.P.P.C.S. (Mains) 2012

Ans. (a & d)

The North-Western region of India receives rainfall in winter due to the western disturbance. Additionally, the coastal areas of Tamil Nadu in South-East India receive rainfall during the winter due to the northeast monsoon.

A – In northwestern India, western disturbance causes rainfall in Punjab, Haryana, Delhi and western Uttar Pradesh.

B – During October and November, the northeast monsoon while crossing over the Bay of Bengal picks up moisture and causes rainfall over Tamil Nadu coast, southern Andhra Pradesh, southeast Karnataka and southeast Kerala.

12. The winter rains caused by Western disturbance in North Western Plain of India gradually decrease from:

- (a) East to West
- (b) West to East
- (c) North to South
- (d) South to North

U.P.P.C.S. (Pre) 2019

Ans. (b)

The winter rains caused by Western Disturbance in the North Western Plain of India gradually decrease from West to East. The extratropical storm originating in the Mediterranean region that brings sudden winter rain to the northwestern parts of the Indian subcontinent is called Western Disturbance. It is a non-monsoonal precipitation pattern driven by the westerlies. Western Disturbance plays an important role in bringing moderate to heavy rain in low-lying areas and heavy snow to mountainous regions of Indian subcontinent, especially in winter. This precipitation has great importance in agriculture, particularly for the Rabi crops.

13. Given below are two statements, in which one is labelled as Assertion (A) and the other as Reason (A) :

Assertion (A) : Rabi crops in India are predominantly cultivated in northwestern states like Punjab and Haryana.

Reason (R) : The success of Rabi crops in these states depends on the rain due to western tropical cyclones during winters.

Select the correct answer using the code given below :

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) (A) is true but (R) is false.
- (c) (A) is false but (R) is true.
- (d) Both (A) and (R) are true but (R) is not the correct explanation on of (A)

U.P. R.O./A.R.O. (Pre) 2023

Ans. (*)

In India, the Rabi crops are harvested primarily in states like Punjab and Haryana in the northwestern region, although Rabi crops are also grown in other parts of India. Rabi crops are winter crops.

They are harvested in northern India, mainly in the north-western region, during the winter season, predominantly in the Indo-Gangetic Plains. Notably, extra Tropical cyclones are steered by the subtropical westerly jet streams, originating from Turkey, Iraq, Iran, Afghanistan and Pakistan. They mostly affect northwestern India (entire northern India) and leads to winter rains, which are essential for Rabi crops.

Both (A) and (R) are true and (R) is the correct explanation of (A). But in this question, tropical cyclone has been given, whereas it is extra tropical cyclone, hence on the basis of this statement, it can be considered wrong. Hence, answer (b) can also be correct.

14. Which one of the following causes is responsible for rainfall during winters in north-western part of India?

- (a) Retreating Monsoon
- (b) Cyclonic depression